Learning Scene Priors From Satellite Imagery for UAV Image Motion Deblurring

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***Abstract*—Uncrewed aerial vehicles (UAVs) are prone to imaging blur due to jitter and environmental disturbances during flight, resulting in degraded image quality. Existing image deblurring methods primarily focus on learning mappings from blurred-clean image pairs, overlooking valuable scene priors in geographically consistent satellite images. To address this limitation, we propose a satellite-guided UAV image deblurring task that leverages the rich information in satellite images to enhance reconstruction quality. First, we present a cross-view UAV deblurring (CVDeblur) dataset consisting of triplet image pairs, with 2092 triplets for training and 4326 for testing. Each triplet includes a blurred UAV image, a geographically consistent satellite image, and a clean UAV image. Subsequently, we propose a dual-branch CVDeblur network (CVDeblurNet), which follows an encode-then-fuse workflow to adaptively integrate satellite features into the UAV feature stream, supplementing texture details. Comparative experiments on CVDeblur show that our CVDeblurNet achieves competitive deblurring performance across scenarios with various blur intensities. Extensive ablation studies validate the beneficial effect of satellite images in UAV image deblurring https://github.com/C-water/CVDeblurNet.**

***Index Terms*—Dataset, drone, dual-branch, image deblurring, satellite.**

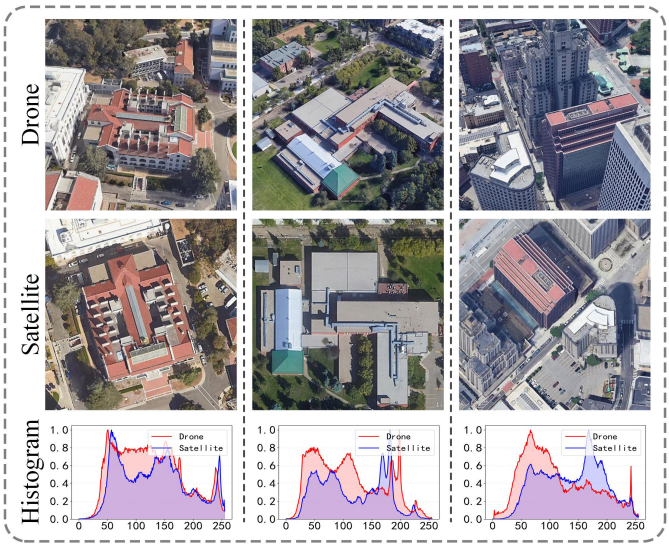


Fig. 1. Comparison of drone images and satellite images with consistent geographical locations.

I. Introduction

With the rapid advancement of uncrewed aerial vehicle (UAV) technology, its applications have been extensively extended to object detection [1], agricultural monitoring [2] and other fields [3], [4]. As a flexible aerial imaging platform, UAVs can efficiently capture large-scale scene information. However, UAV imaging is easily affected by airflow turbulence and fuselage vibration, which cause motion blur and seriously degrade the accuracy of subsequent visual analysis. Therefore, UAV image motion deblurring has become an essential pre-processing step for high-quality aerial vision applications. Most commercial UAVs adopt hardware-based solutions, such as optical image stabilization and sensor-shift compensation. In contrast, post-processing algorithms require no hardware modifications and offer the advantages of low cost and high flexibility. Such strategies have been well validated in general image restoration [5], [6], [7], [8], and serve as a practical technical approach for UAV image deblurring.

Over the past decade, deep learning has significantly advanced general-purpose image deblurring [9], [10], [11], [12], [13], [14], [15], [16], [17]. Recent methods have achieved promising performance on public benchmarks, such as GoPro [18] and MC-Blur [19], which mainly model general motion patterns and natural image priors from ground-view scenes. Nonetheless, these methods cannot be directly applied to UAV images due to inherent domain gaps. UAV imaging adopts a top-down perspective, leading to distinct object geometry, wider field of view, lower per-object resolution, and weaker texture details compared with ground-view images. These differences severely limit the direct deployment of models trained on ground-view data for UAV image deblurring.

Alongside single-image deblurring, reference-based image restoration has emerged as a promising paradigm. It compensates for missing details in degraded inputs by leveraging high-quality reference images with consistent semantic content, and has been successfully used in super-resolution, denoising, and deblurring. In remote sensing, satellite imagery provides globally sharp, low-noise, and geographically aligned observations of the same ground scene captured by UAVs.

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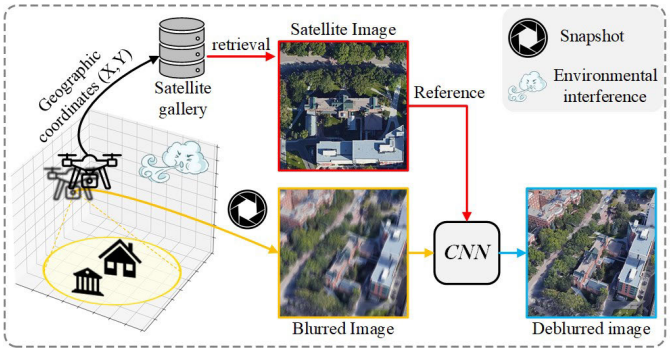


Fig. 2. Schematic of the deblurring process for UAV images guided by satellite images.

Despite differences in imaging time, sensors, resolution, and viewpoint, satellite and UAV images share consistent spatial structure, terrain, and architectural layout (see Fig. 1). Such unique advantages make satellite images an ideal and easily accessible external prior for UAV image restoration. However, there is still a lack of dedicated datasets and customized network designs that combine cross-view scene priors with UAV motion deblurring.

As shown in Fig. 2, we introduce a novel satellite-guided UAV image deblurring task, aiming to enhance restoration quality by exploiting scene priors from satellite images. To support this research, we develop a dedicated dataset along with a baseline model. First, we construct a triplet-structured cross-view UAV deblurring (CVDeblur) dataset, which mainly consists of UAV blurred images, UAV sharp images, and satellite-view images. This dataset adopts nonoverlapping scenes between the training and test sets to prevent evaluation bias caused by scene repetition. The training set contains 2092 triplets with moderate blur intensity to promote learning of common blur patterns. The test set comprises 1402×3 triplets subdivided into low, middle, and high blur-intensity subsets, enabling comprehensive assessment of model generalization under varying degradation levels. Besides the above data, it also includes real-world samples composed of 120 pairs of blurred UAV images and satellite reference images. Additionally, CVDeblur includes ground-view image pairs (as auxiliary data) to investigate the impact of view differences on UAV image deblurring performance. The seven usage configurations of CVDeblur are illustrated in Fig. 3.

Second, we design a baseline network named CVDeblur network (CVDeblurNet), which comprises a drone branch and a satellite branch. The network takes a blurred UAV image and a satellite reference image as dual inputs, following an encode-then-fuse strategy to integrate cross-view information. The satellite branch employs four single-input encoders (SinE) to extract high-level features from the satellite image. The features output by the last two SinE are injected into the drone branch. Similarly, the drone branch first adopts two SinE to encode the blurred UAV image, and then feeds both UAV features and satellite features into two double-input encoders (DouE) for feature selection and fusion. Finally, a 3×3 convolutional layer combined with residual connections is utilized to reconstruct the sharp UAV image. Experiments on the CVDeblur dataset demonstrate that CVDeblurNet achieves competitive performance compared with 17 learning-based methods, while extensive ablation studies verify the potential gains provided by satellite images.

In summary, the main contributions of this article include as follows.

1) We introduce a novel task of satellite-guided UAV image deblurring.
2) We construct the CVDeblur dataset composed of synthetic and real-world samples, including blurred UAV images and clean versions, as well as geographically consistent satellite images.
3) We design the CVDeblurNet, which exploits scene information from satellite images to improve UAV image deblurring performance.
4) Extensive experiments demonstrate that CVDeblurNet extracts valuable information contained in satellite images and achieves superior performance.

The remainder of this article is structured as follows. We first introduce related works in Section II. In Section III, we present the proposed CVDeblur dataset in detail. Then, we introduce the CVDeblurNet in Section IV. The experimental results are presented in Section V. Finally, the concluding remarks are drawn in Section VI.

II. Related Work

A. Single Image Deblurring

Single-image deblurring is a foundational task in computer vision, which aims to recover sharp and high-fidelity images from blurred versions. Most traditional image deblurring methods [20], [21], [22], [23], [24] model motion deblurring as an optimization problem and estimate a blur kernel that characterizes the image blurring process. However, these methods often struggle with complex real-world blurs due to the limited representational capacity of handcrafted priors.

The rise of deep learning has transformed single-image deblurring, allowing for the data-driven learning of intrinsic blur patterns and image priors. The key to learning-based methods lies in designing a suitable neural network structure. For instance, Nah et al. [18] introduced a multiscale convolutional neural network (CNN) that progressively refined deblurred results. Kupyn et al. [25] proposed the DeblurGAN, which employed a generative adversarial network (GAN) [26] to capture perceptual realism. Tao et al. [27] developed a recursive residual network (RRN) [28] to reduce computational costs without sacrificing accuracy. Zhang et al. [29] proposed a dual-GAN deblurring framework consisting of BGAN and DBGAN, which learns realistic blurring via a relativistic blur loss. These learning-based methods have achieved remarkable progress on standard benchmarks (e.g., GoPro [18], REDS [30], HIDE [31], and MC-Blur [19]).

Motion deblurring for UAV images, as a crucial branch of single-image deblurring, faces unique challenges arising from platform instability (e.g., jitter, sudden motion) and complex environmental interference. Due to differences in imaging perspectives and blur patterns, models trained on ground-view data achieve limited performance when processing UAV

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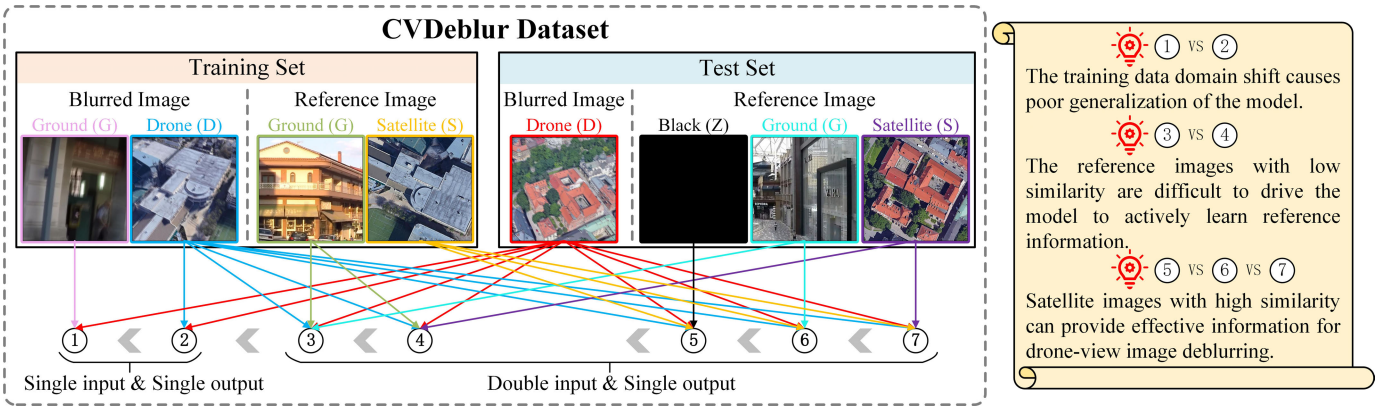


Fig. 3. Schematic of seven usage strategies for the CVDeblur dataset.

images and suffer from pixel misalignment. Consequently, several studies have initiated targeted exploration into UAV-specific blur problems. Existing works [11], [32], [33], [34], [35], [36] have attempted to adapt mechanisms commonly used in general image deblurring, such as GAN and attention mechanisms [37], [38], [39], to the task of UAV image deblurring. Recently, Tong et al. [40] proposed a benchmark of a real UAV infrared image motion blur dataset, U2IRD. Unlike U2IRD, we focus on color images and present the CVDeblur dataset and the benchmark comprising 15 methods.

B. Reference-Based Image Restoration

The reference-based image restoration task focuses on restoring degraded target images guided by high-quality reference images that share similar content. Early methods primarily relied on hand-crafted similarity metrics, such as SIFT [41] and patch matching [42]. These methods accomplish the restoration by aligning and transferring textures from reference images to target images. However, these methods struggle to handle significant appearance variations or complex degradation patterns. With the advancement of deep learning, data-driven methods have significantly enhanced the performance of reference-based image reconstruction. Early research endeavors concentrated on reference-based image super-resolution and developed diverse techniques for selecting and fusing relevant features from input and reference images [43], [44], [45], [46]. For instance, Zheng et al. [47] proposed a cross-scale warping method to achieve non-rigid image transformation. They also adopted FlowNetS [48] to establish multiscale correspondences between LR and HR images. Jiang et al. [49] proposed a dual contrastive matching method C^2-Matching to address both transformation and resolution discrepancies between input and reference images. Recent studies [50], [51], [52], [53] have shifted focus to more diverse degradation types such as noise and blur, thereby further verifying the auxiliary value of reference images in the image restoration process. For instance, Zhang et al. [50] proposed a three-stage reference-based restoration framework. It reconstructs high-quality output images through sequential preliminary restoration of the input image and transfer of effective information from the reference image.

A key aspect of reference-based image restoration research lies in constructing similar target-reference image pairs. The currently prevalent methods entail collecting reference images based on the principle of spatial consistency to ensure visual similarity. Reference images can be obtained either by selecting from image sequences or synchronous capture via multicamera systems. For example, Luo et al. [54] addressed the image defocus deblurring task and employed synchronously captured ultra-wide-angle images with similar content as reference images. In the remote sensing field, the collection of spatially consistent satellite images is relatively straightforward owing to standardized satellite image libraries and accurate latitude-longitude information. Consequently, several studies have explored satellite image super-resolution with historical satellite images serving as [45] and [55]. UAVs record latitude and longitude information in real time, which enables the accurate acquisition of satellite images with consistent geographic locations. Building upon this, we propose the satellite image-guided UAV image deblurring task, aiming to enhance the reconstruction quality of UAV images by leveraging the structural and textural information contained in satellite images. Due to the constraints from multidimensional differences, including visual sensors, imaging perspectives, imaging times, and other aspects, this task is rendered highly challenging.

III. CVDeblur Dataset

In this section, we first elaborate on the differences between the CVDeblur dataset and existing datasets. Next, we introduce the key features of the CVDeblur dataset and the corresponding build details.

A. Dataset Overview

As summarized in Table I, existing RGB image deblurring datasets primarily focus on ground-view images and lack reference images. Several datasets focus on deblurring UAV infrared images. In contrast, our CVDeblur dataset focuses on UAV RGB image deblurring and incorporates satellite images as reference. To enhance dataset diversity, we introduce three levels of motion blur with varying intensities into the synthetic test set, resulting in 1402×3 test images. We also collect real-world samples to validate the effectiveness of the proposed method. Furthermore, t-SNE visualization in Fig. 4 reveals

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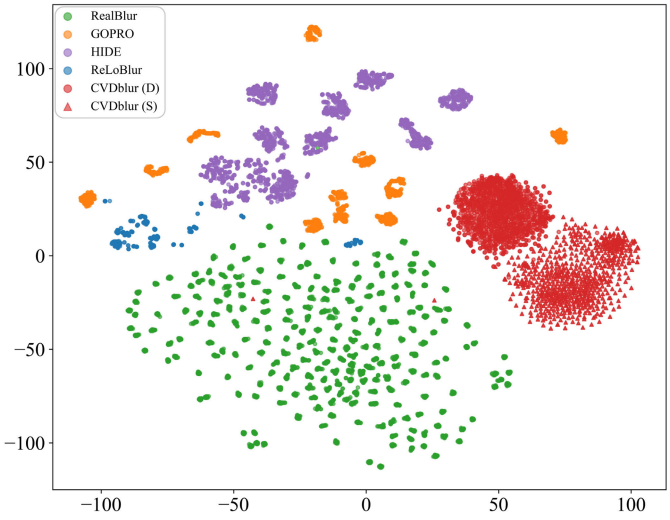


Fig. 4. Visualization of various datasets using t-SNE.

TABLE I

REPRESENTATIVE BENCHMARK DATASETS FOR IMAGE MOTION DEBLURRING. IMAGING VIEW: G = GROUND, D = DRONE, AND S = SATELLITE

Datasets	Reference -based	View	Average Resolution	Image Number	
				Training set	Test set
GoPro [18]	✗	G	1280×720	2103	1111
RealBlur [56]	✗	G	668×760	3758	980
HIDE [31]	✗	G	1280×720	6397	2025
RsBlur [57]	✗	G	1916×1200	8878	4480
ReLoBlur [58]	✗	G	2152×1436	2010	395
SBTI* [59]	✗	D	640×480	5,600	1,400
UIRD* [60]	✗	D	640×512	25,282	7,885
U2IRD* [40]	✗	D	640×480	1459	365
CVDeblur	✓	D+S	512×512	2092	1402×3+120

* denotes the dataset consists of infrared image.

a distinct distribution gap between UAV and ground-view images, while satellite images and UAV images exhibit a close distributional distance.

B. Dataset Construction

Fig. 5 illustrates the construction workflow of the CVDeblur dataset, including the drone- and satellite-view images as well as other relevant ground-view images. Table II presents the statistical information of the CVDeblur dataset.

1) Drone and Satellite Image Collection: There are two key steps in constructing the CVDeblur dataset: 1) collecting paired UAV and satellite images and 2) adding motion blur to UAV images. To obtain geographically aligned satellite and drone images, we choose University-1652 [61], a representative dataset in cross-view geo-localization, as the image source. Each scene in University-1652 contains ground-view (discarded in this work), drone-view, and satellite-view images, which satisfy our requirements. Specifically, each scene includes 54 drone images and one satellite image, with nonoverlapping scenes between the training set and test set. Thus, we randomly select 2–4 drone images from each training scene and two drone images from each testing scene, resulting in a total of 3494 drone images (2092 for training and 1402 for testing). The satellite image from each scene serves as the corresponding reference image. Subsequently, we followed the DeblurGAN [25] to add motion blur to drone images. Specifically, a 2-D random motion trajectory is generated using a Markov process, where the displacement at each time step is determined by the current velocity, an inertia term, a Gaussian disturbance, and an impulse disturbance. Notably, for the test set, we vary the "*kernel length*" parameter to randomly generate blur kernels with three distinct strength ranges, which serves to verify the model's generalization ability in handling unseen blur intensities during training. Fig. 6 illustrates three levels of blur kernels. In addition, we capture blurred UAV images with the DJI Mini3 device. Leveraging the built-in latitude and longitude information embedded in the captured UAV images, we collect geographically matched satellite images via the Google Earth Pro platform. The real-world test set composed of 120 image pairs is constructed, with partial real image pairs presented in Fig. 7.

2) Other Image Collection: The CVDeblur dataset requires verification of the incompatibility between ground- and drone-images, in both training data and reference images. Thus, we extend the CVDeblur dataset from three aspects, as illustrated in Fig. 5

1) We introduce ground-view images into the training set as input and reference images, to evaluate the generalization capability of models trained on ground-view data. Specifically, we randomly crop 2092 pairs of 512×512 clean-blurred images from the GoPro dataset as auxiliary training samples, and simultaneously select another 2092 clean images as reference images.
2) We augment the synthetic test set with ground-view reference images to verify the impact of reference image selection during testing. We randomly crop 1402 images of 512×512 resolution from the RealBlur and HIDE datasets as reference images.
3) We add 1402 pure black images to the CVDeblur synthetic test set to verify performance degradation under the no-reference-image setting.

IV. METHODS

A. Overall Network Architecture

The overall structure of CVDeblurNet, presented in Fig. 8, consists of a drone branch and a satellite branch. Concretely, the satellite image $I_{ref} \in \mathbb{R}^{H\times W\times 3}$ is first embedded by a 3×3 convolutional layer to produce an initial feature representation $f^0_{ref} \in \mathbb{R}^{H\times W\times C}$. This feature is then fed into four stacked SinE. Each SinE comprises two multiscale interaction blocks with a single input (MIB-S), a 3×3 convolution layer, and a residual connection. The output feature of MIB-S in the last two SinE is fed into the drone branch as auxiliary information. The overall process of the satellite branch is summarized as

$$\begin{cases} f^0_{ref} = Conv_{3\times 3}\left(I_{ref}\right) \\ f^i_{ref} = SinE^i\left(f^{i-1}_{ref}\right), \quad i = 1, 2, 3, 4. \end{cases} \tag{1}$$

For the UAV image $I_{in} \in \mathbb{R}^{H\times W\times 3}$, it is first mapped to high-dimensional feature $f^0 \in \mathbb{R}^{H\times W\times C}$ via a 3×3 convolutional layer. Subsequently, four stacked encoders are also applied to process UAV features. Unlike the satellite branch, the encoders in the drone branch include two types: 1) SinE and 2) DouE. The inputs of DouE consist of the UAV feature from the

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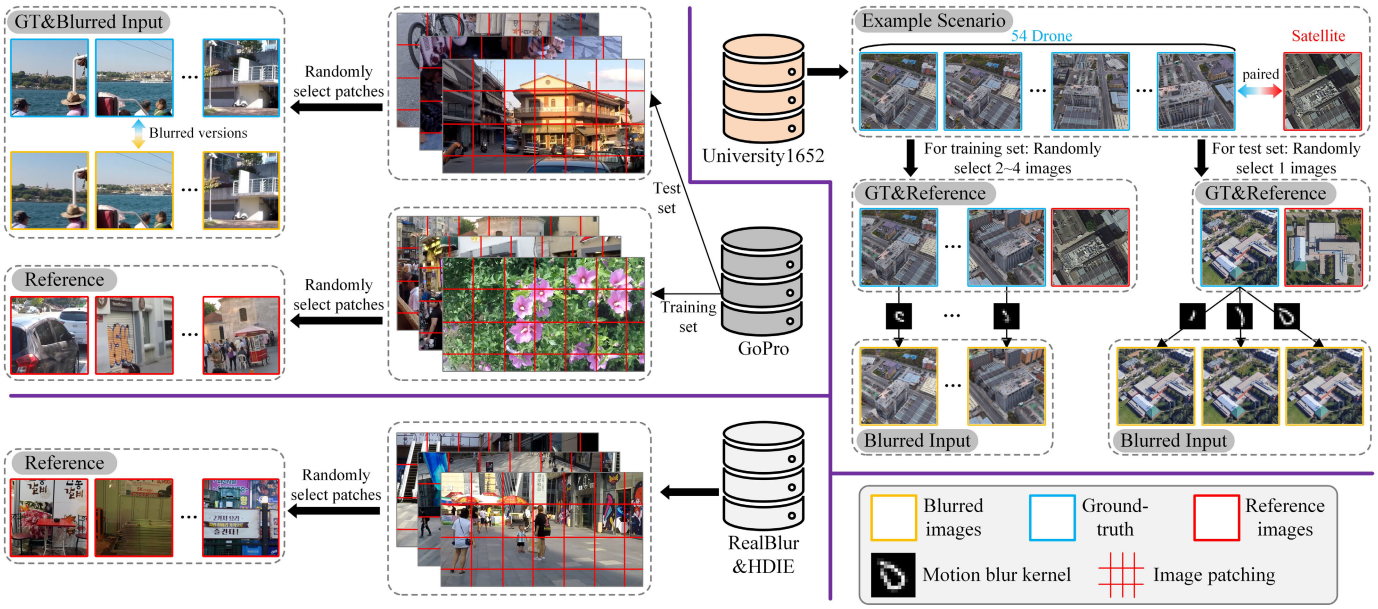


Fig. 5. Construction workflow of the CVDeblur dataset, including the core drone- and satellite-view images as well as other relevant ground-view images.

TABLE II

SUMMARY OF CVDEBLUR DATASETS. IMAGES IN CVDEBLUR ARE COLLECTED FROM UNIVERSITY1652 [61], GOPRO [18], REALBLUR [56], AND HIDE [31] DATASETS. WE ALSO ACQUIRED ADDITIONAL REAL-WORLD DATA USING DJI MINI3 DEVICE

Data types	Training set				Synthetic test set				Real test set	
	Main image		Reference images		Main images	Reference images			Main images	Reference images
View	Ground	Drone	Ground	Satellite	Drone	Black	Ground	Satellite	Drone	Satellite
Number	2092	2092	2092	2092	1402×3	1402	1402	1402	120	120
Source	Gopro	University1652	Gopro	University1652	University1652	–	RealBlur&HDIE	University1652	–	–

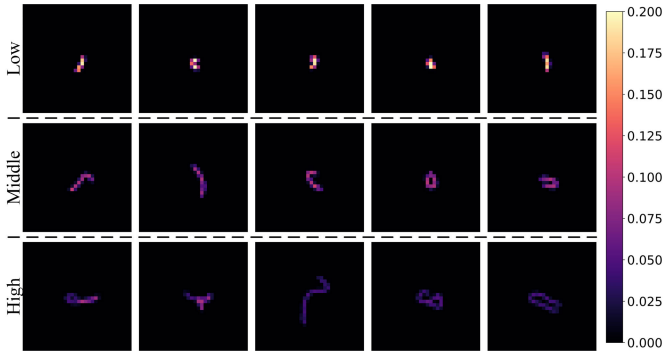


Fig. 6. Visualization of blur kernels with three different intensities. The trajectory length reflects the degree of blur, and the total pixel value of a single blur kernel is 1.

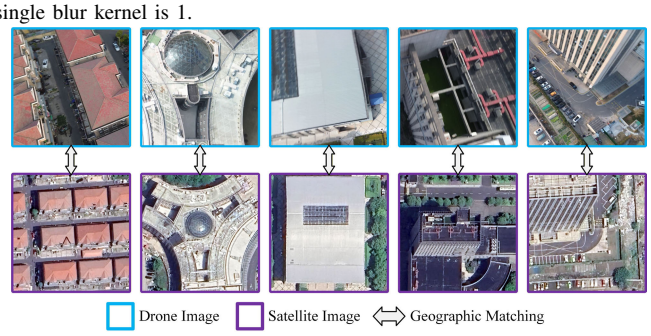


Fig. 7. Paired UAV-satellite samples with consistent geographic locations.

previous encoder and the auxiliary feature from the satellite branch at the same stage, which are fused via a multiscale interaction block with double inputs (MIB-D). Finally, the output feature of the last DouE is processed by a 3×3 convolutional layer and a residual connection to reconstruct the sharp UAV image I_{out}. The overall process of the drone branch is summarized as

$$\begin{cases} f^0 = Conv_{3\times3}\left(I_{in}\right) \\ f^i = SinE^i\left(f^{i-1}\right), \quad i = 1,2 \\ f^i = DouE^{i-2}\left(f^{i-1}, f_{ref}^{i-1}\right), \quad i = 3,4 \\ I_{out} = Conv_{3\times3}\left(f^4\right) + I_{in}. \end{cases} \tag{2}$$

We optimize the proposed network using the L1 loss

$$\mathcal{L}\left(I_{out}, I_{gt}\right) = \frac{1}{n}\sum_{i=1}^{n}\left|I_{out}^i - I_{gt}^i\right|. \tag{3}$$

Next, we describe each block of our CVDeblurNet.

B. Network Modules

1) MIB With Single/Double Input: UAV and satellite images are both categorized as bird's-eye-view (BEV) images, featuring rich scene information and variable fields of view. On the basis of extracting local features, integrating large receptive fields to learn global structural information facilitates the restoration of sharp details. Thus, we design a multiscale interaction block (MIB) that balances receptive field size and computational efficiency. The MIB is further divided into MIB-S and MIB-D based on the number of input features. As shown in Fig. 9, the MIB consists of six parallel context learning blocks (PCLBs), four single-domain or dual-domain

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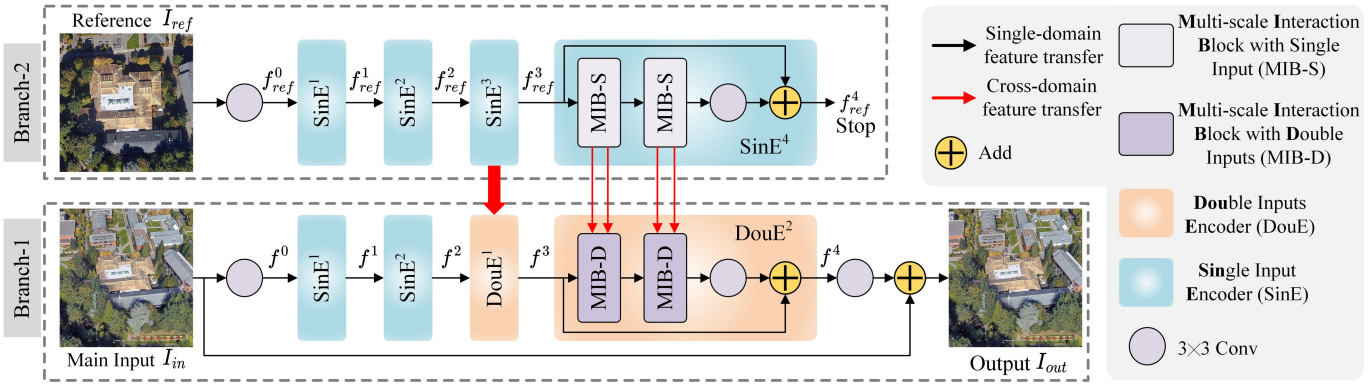


Fig. 8. Overall architecture of the proposed CVDeblurNet, consisting of a drone branch (Branch-1) and a satellite branch (Branch-2).

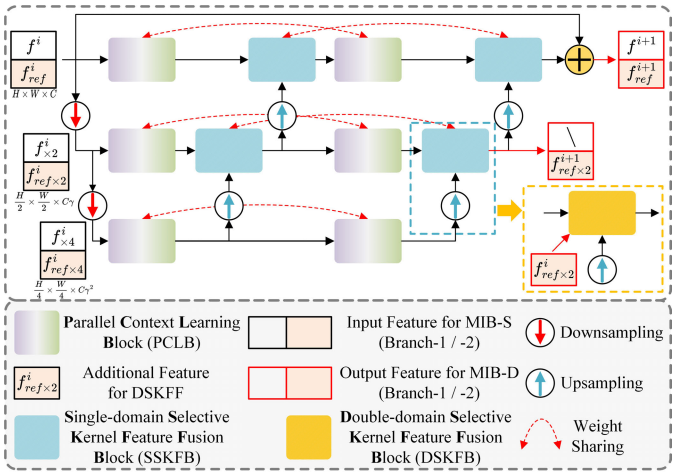


Fig. 9. Network architecture of the MIB-D/MIB-S.

selective kernel feature fusion blocks (SSKFB/DSKFB). These components, along with downsampling and upsampling layers, are organized across three scale levels. Details of the MIB-S and MIB-D variants are provided in the next paragraph.

The MIB-S follows the same processing flow for both Branch-1 and Branch-2. Taking Branch-2 as an example, the input reference feature f_{ref}^i is first downsampled twice to generate a multiresolution feature set $\{f_{ref}^i,\ f_{ref\times 2}^i,\ f_{ref\times 4}^i\}$. Then, features at each resolution are fed into PCLB for contextual information enhancement, and subsequently fused via SSKFB, which adaptively integrates features from the same and lower scales. For MIB-D in DouE, it takes UAV image features and satellite image features as dual inputs. Compared with MIB-S, its key difference lies in replacing the four SSKFBs with triple-input DSKFBs to promote cross-view image fusion, as illustrated by the yellow dashed box at the bottom right of Fig. 9. Inspired by prior work [62], a weight-sharing mechanism is also adopted for MIB-S and MIB-D, enabling PCLB and SSKFB/DSKFB to perform secondary feature refinement without increasing parameters.

2) Parallel Context Learning Block: In addition to employing a multiscale structure to enlarge receptive fields, we design a PCLB to model global dependencies, as shown in Fig. 10. The PCLB contains parallel channel and spatial enhancement components. In contrast to conventional channel or spatial attention mechanisms that capture feature correlations along a single dimension, our PCLB generates attention features while preserving correlations across spatial and channel dimensions. Given an input feature $f_{in} \in \mathbb{R}^{H\times W\times C}$, two group convolution layers combined with an activation layer are employed to obtain the feature $f' \in \mathbb{R}^{H\times W\times C}$. Then, this feature is divided into two parts (f_c^{in} and f_s^{in}) along the channel dimension, each with a shape of $H \times W \times C/2$, and fed into the channel and spatial enhancement components, respectively.

For channel enhancement, we employ matrix multiplication to construct the global channel feature $f_c' \in \mathbb{R}^{1\times 1\times C/2}$, while establishing correlations between the spatial and channel dimensions. The construction of feature f_c' is formulated as

$$f_c' = \mathcal{R}\left(f_c^{in}\right) \times \mathcal{S}\left(\mathcal{R}\left(Conv_{1*1}\left(f_c^{in}\right)\right)\right) \tag{4}$$

where $\mathcal{R}(\cdot)$ and $\mathcal{S}(\cdot)$ denote reshape and softmax operations. The global channel feature f_c' is then aggregated to each position of the input feature f_c^{in} through element-wise addition, yielding the channel-enhanced feature $f_c \in \mathbb{R}^{H\times W\times C/2}$.

For spatial enhancement, a similar strategy is adopted to generate the global spatial feature $f_s' \in \mathbb{R}^{H\times W\times 1}$. The generation process is formulated as

$$f_s' = \mathcal{R}\left(\mathcal{R}\left(f_s^{in}\right) \times \mathcal{S}\left(GAP\left(Conv_{1*1}\left(f_s^{in}\right)\right)\right)\right). \tag{5}$$

This feature is aggregated element-wise to f_s^{in} to produce the spatial-enhanced feature $f_s \in \mathbb{R}^{H\times W\times C/2}$.

Finally, two enhanced features are expanded to the same dimension via a convolution layer and then fused, resulting in the output feature $f_{out} \in \mathbb{R}^{H\times W\times C}$ of the PCLB

$$f_{out} = f_{in} + \sigma\left(Conv_{1*1}\left(f_c'\right)\right) + \sigma\left(Conv_{1*1}\left(f_s'\right)\right) \tag{6}$$

where $\sigma(\cdot)$ represents an activation layer.

3) Single/DSKFB: A key objective of CVDeblurNet is to fuse satellite features with UAV features, further enhancing the quality of UAV image reconstruction. Inspired by the selective kernel fusion mechanism [63] for fusing homogeneous features, we develop a double-domain selective kernel feature fusion block (DSKFB), as illustrated in Fig. 11.

The DSKFB accepts three inputs: UAV feature and satellite feature at the same scale, along with upsampled lower-scale UAV feature, denoted as z_{up}, z_{ref}, and z_{low}. It first constructs a global feature descriptor $v \in \mathbb{R}^{1\times 1\times C/8}$ as follows:

$$v = \sigma\left(Conv\left(GAP\left(z_{up} + z_{ref} + z_{low}\right)\right)\right). \tag{7}$$

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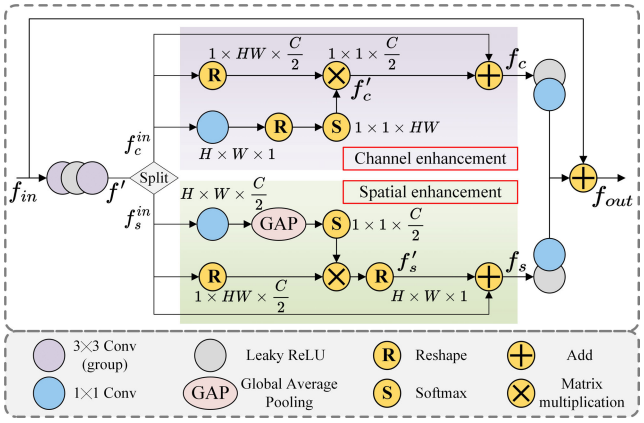


Fig. 10. Network architecture of the PCLB.

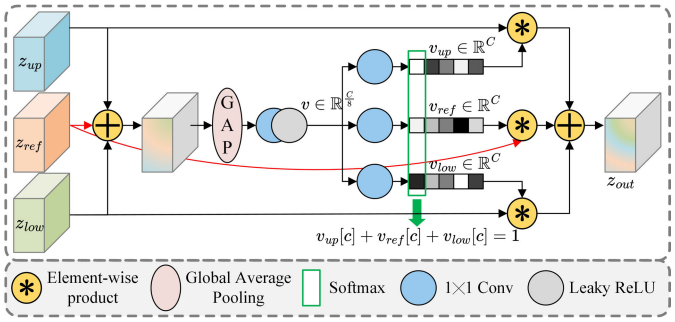


Fig. 11. Network architecture of the DSKFB.

Subsequently, three independent convolutional layers are applied to estimate the importance of each input feature

$$v_{up} = Conv_{up}(v), v_{ref} = Conv_{ref}(v), v_{low} = Conv_{low}(v) \quad (8)$$

where v_{up}, v_{ref}, and v_{low} correspond to the estimated weights for the input features z_{up}, z_{ref}, and z_{low}, respectively. To balance the weights of features from different views, the softmax function is applied to normalize these three weights. Note that the SSKFB comes from [62], which is equivalent to removing all components related to reference features in DSKFB. Although SSKFB and DSKFB share a similar structure, they serve distinct functional purposes. The DSKFB performs channel-wise adaptive selection on cross-view features to retain feature maps contributing to UAV image deblurring.

V. Experiments and Results

A. Evaluation Metrics and Implementation Details

1) Evaluation Metrics: Following previous image deblurring methods, peak-signal-to-noise ratio (PSNR) and structure similarity (SSIM) [81] are employed for evaluating model performance. We also adopt the number of learnable parameters (Params) and multiply-accumulate operations (MACs) to measure the computational cost of models. Note that the MACs are measured by processing a 512 × 512 image.

2) Implementation Details: For CVDeblurNet, the number of channels of the three-scale features in the MIB within each encoder increases sequentially as resolution decreases. The corresponding channel counts are 48, 48γ, and 48γ^2, with γ set to 1.5. We adopt the cosine annealing strategy [82] and Adam

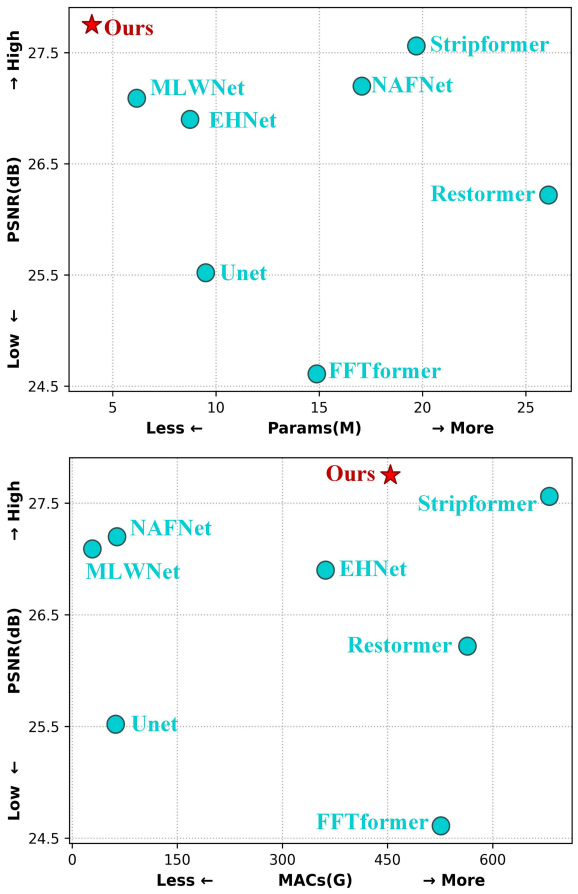


Fig. 12. PSNR versus computational costs (parameters and MACs).

[83] optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The learning rate is initialized as 1×10^{-3}, and decayed by 0.75 after 50 epochs. We set the batch size as 1 and the input patch size as 512×512. The total number of epochs is 350. All experiments are conducted using PyTorch on an NVIDIA RTX 3090 GPU with 24 GB of memory.

B. Comparison With State-of-the-Art

In this section, we present comparative experiments to evaluate the effectiveness of the proposed CVDeblurNet. The experiments are divided into two parts. First, we select four pre-trained methods based on the GoPro dataset and test them on the CVDeblur dataset. Second, we select ten methods to retrain and test them on the CVDeblur dataset.

As shown in Table III, the quantitative results of GoPro pretrained methods are generally lower. The average PSNR values of the optimal XYScanNet model in this group are 2.36 dB lower than CVDeblurNet on the three test sets. This verifies that the domain shift between datasets hinders the generalization of pretrained models to UAV images. Additionally, comparisons among the ten retrained models and CVDeblurNet show that the proposed method achieves competitive results across all metrics and obtains the optimal performance in terms of PSNR. This demonstrates the effectiveness of the CVDeblurNet architecture. As shown in Fig. 12, the comparison of parameters also indicates that CVDeblurNet maintains a reasonable computational burden. Although our

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TABLE III

PERFORMANCE COMPARISON OF OUR CVDEBLURNET AND OTHER RELATED WORKS ON CVDEBLUR DATASET. THE TEST RESULTS ARE DIVIDED INTO FOUR CATEGORIES: LOW, MIDDLE, HIGH, AND AVERAGE. BEST RESULTS ARE BOLDFACED. ALL RETRAINED MODELS ADOPT THE SAME LEARNING RATE AND ITERATION NUMBER

Methods	Low		Middle		High		Average		Computational cost	
	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	Params (M)	MACs (G)
Input	19.16	0.444	19.72	0.467	19.61	0.445	19.50	0.452	–	–
M^2AENet† [64]	26.46	0.852	25.43	0.816	24.16	**0.756**	25.35	0.808	13.41	49.06
MRDNet† [65]	26.37	0.852	24.75	0.785	23.29	0.700	24.80	0.779	12.19	383.09
XYScanNet† [66]	26.70	0.863	25.57	0.824	23.92	0.746	25.39	0.811	14.85	484.85
EVSSM† [67]	26.40	0.858	25.49	0.822	23.90	0.755	25.26	0.812	16.26	715.98
FFTformer* [68]	24.35	0.819	26.74	0.870	22.73	0.660	24.61	0.783	14.88	526.14
MIRNet* [69]	24.81	0.812	26.83	0.869	22.91	0.665	24.85	0.782	31.78	3265.00
Restormer* [70]	26.41	0.857	28.78	0.895	23.48	0.687	26.22	0.813	26.09	563.96
Concertormer* [71]	26.23	0.848	27.51	0.871	23.01	0.666	25.58	0.795	26.39	106.56
LoFormer* [72]	25.95	0.850	26.62	0.856	23.27	0.686	25.28	0.797	16.40	52.19
EHNet* [73]	28.66	0.920	28.29	0.896	23.76	0.700	26.90	0.838	8.74	361.45
MB-TylorFormerv1** [74]	29.11	0.922	28.75	0.904	24.35	0.731	27.40	0.852	1.26	61.59
MB-TylorFormerv2** [75]	28.08	0.909	28.75	0.905	**24.49**	0.729	27.10	0.847	1.384	51.886
Unet [76]	26.49	0.878	26.87	0.866	23.20	0.685	25.52	0.810	9.50	61.95
NAFNet [77]	28.23	0.909	29.30	**0.919**	24.08	0.713	27.20	0.847	17.06	63.87
MPRNet [78]	27.88	0.899	29.36	0.916	23.88	0.705	27.04	0.840	**3.63**	2195.00
Stripformer [79]	29.08	0.926	29.37	0.918	24.23	0.725	27.56	0.856	19.70	680.73
MLWNet-T [80]	27.94	0.899	29.13	0.914	24.21	0.717	27.09	0.843	6.16	**28.78**
Ours	**29.28**	**0.926**	**29.53**	0.918	24.45	0.731	**27.75**	**0.858**	3.99	454.10

† denotes that the models adopt pre-trained weights based on the GoPro dataset. Due to differences in blur patterns, these models exhibit spatial misalignment leading to lower PSNR. We manually adjust pixel alignment to achieve the optimal PSNR.

* denotes the model is trained with 256×256 images as input due to computational resource constraints.

** indicates that the model structure is adjusted to take 512×512 images as input. For MB-TylorFormerv1, the basic channel number is set to 16 (originally 24). For MB-TylorFormerv2, the basic channel number is adjusted to 14 (originally 28), and the number of modules is configured as [2,2,2,2] (originally [4,6,6,8]).

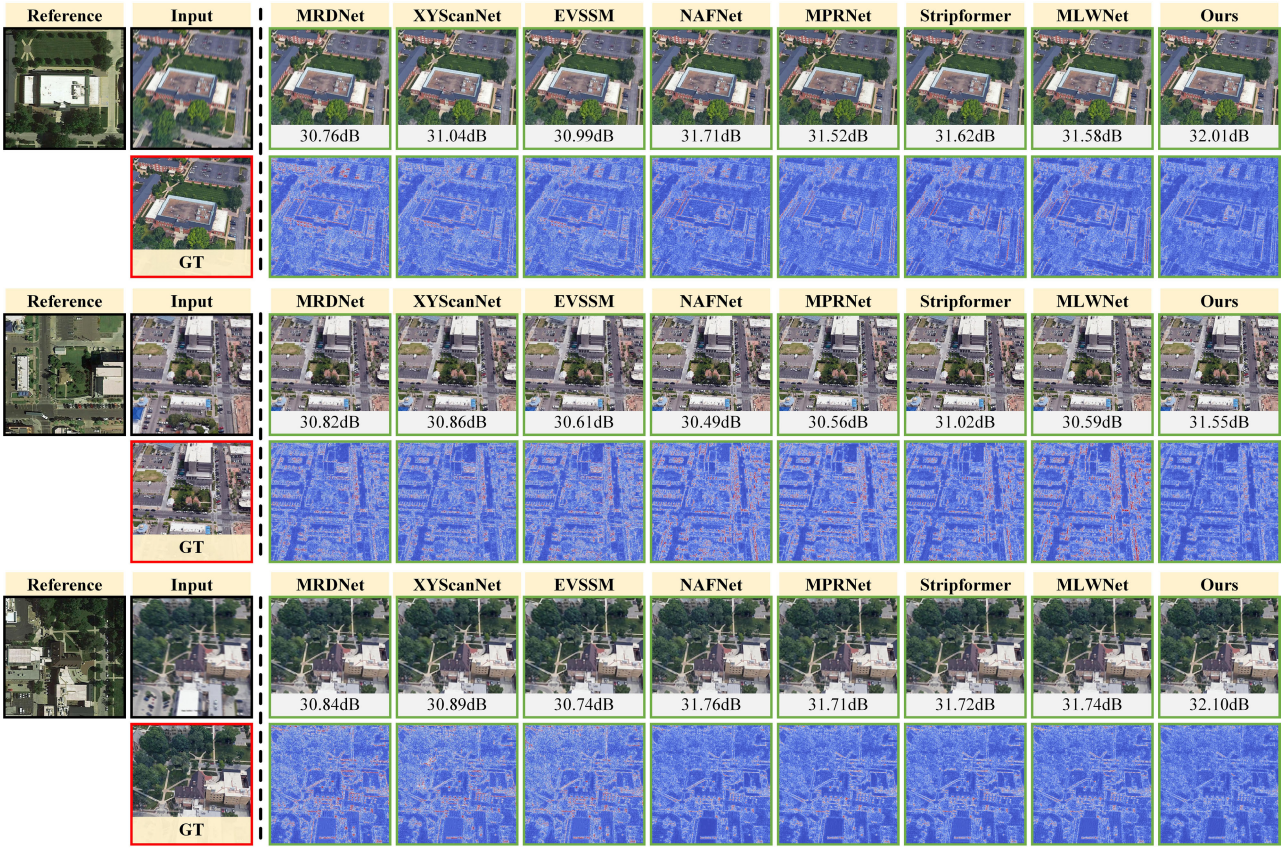


Fig. 13. Deblurred results by different methods on the images from the CVDeblur dataset.

model has a relatively small number of parameters, the multiscale structure (×1, ×2, ×4) in the MIB leads to relatively high MACs. In contrast, MLWNet exhibits distinct advantages in both parameters and MACs, validating that deblurring in the frequency domain is a feasible and effective strategy. Such merits will be absorbed and leveraged in our future research to design more efficient deblurring models.

Fig. 13 presents restoration results of samples with varying blur intensities. To highlight performance differences among various methods, we calculate corresponding error maps for

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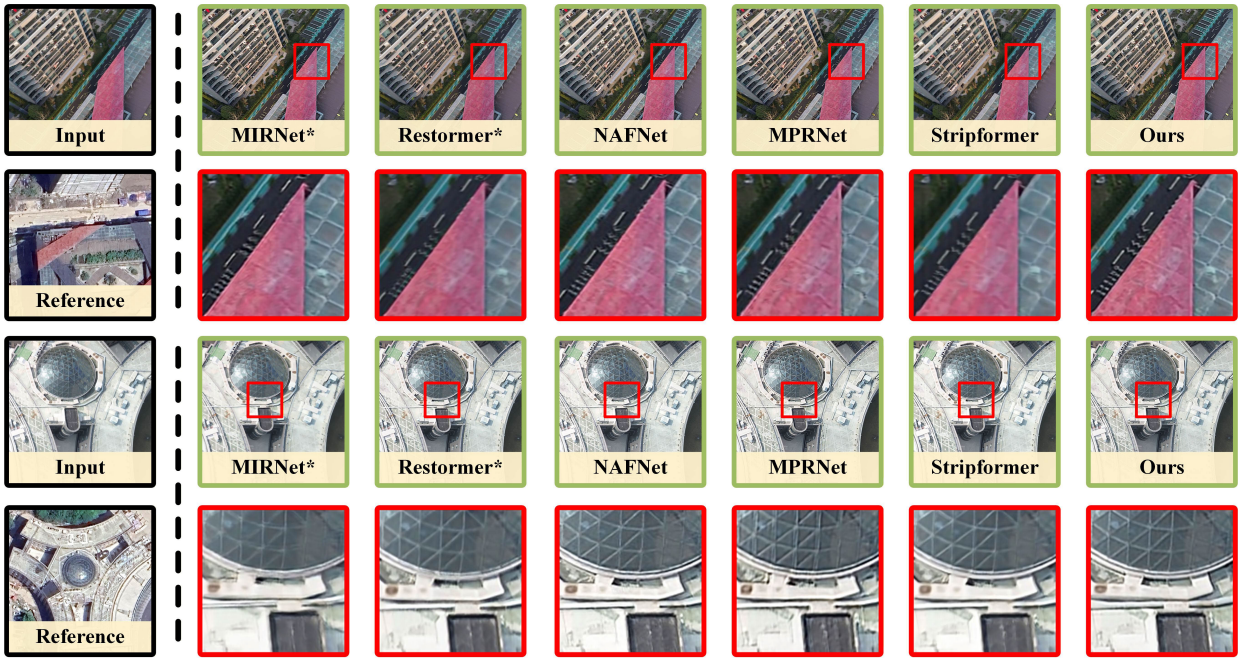


Fig. 14. Deblurred results by different methods on the images from the real test set.

TABLE IV
PERFORMANCE COMPARISON ON THE REAL TEST SET

Methods	MUSIQ↑	MANIAQ↑	CLIPIQA↑
Input	54.6659	0.2223	0.2605
MIRNet*	62.5466	0.2877	0.4056
Restormer*	55.8155	0.2418	0.3334
NAFNet	57.0627	0.2796	0.3786
MPRNet	62.9586	**0.3021**	0.4421
Stripformer	61.7876	0.3005	0.4549
Ours	**64.4107**	0.2997	**0.4621**

each method. It can be observed that CVDeblurNet consistently exhibits smaller errors in fine edge details for input images with different blur intensities.

C. Performance Evaluation on Real-World Data

We perform comparative experiments on the real test set using models pre-trained on synthetic data. Unsupervised image evaluation metrics, including MUSIQ [84], MANIQA [85], and CLIPIQA [86], are adopted to evaluate the quality of output images. Results are summarized in Table IV, from which it can be observed that our method achieves top-three performance across all quantitative metrics and delivers the optimal overall performance. Visualization results are provided in Fig. 14, where detailed regions are enlarged for clearer observation. These visual comparisons demonstrate the advantages of our method in image detail restoration.

D. Ablation Studies

To verify the effectiveness of CVDeblurNet, we conduct extensive ablation studies, including

1) To verify the feasibility of satellite image-guided UAV image deblurring.
2) To explore the impact of satellite image variations on the model performance.
3) To evaluate the contribution of individual components of CVDeblurNet.
4) To examine the impact of the number of encoders and channels on performance.

1) Effectiveness of Satellite Images: To evaluate the contribution of satellite images to UAV image deblurring, we conduct a comparative study by varying the reference image settings during training and testing. As summarized in Table V, in addition to the full CVDeblurNet (#7 model), six models (#1∼#6 models) are trained and evaluated under different reference image configurations. The following observations can be drawn.

1) When only the drone branch of CVDeblurNet is utilized, the #1 model trained on ground-view images exhibits significantly poorer performance on UAV data, showing an average PSNR decrease of 8.35 dB relative to the #2 model trained on UAV images. This result indicates that variations in blur patterns introduced by viewpoint differences substantially limit model generalization, even under equal training data conditions.
2) Building upon the #2 model, #3 and #4 models incorporate ground-view reference images irrelevant to UAV image content during training, which instead reduces image restoration quality. Furthermore, comparison between #3 and #4 models shows that using reference images inconsistent with those in training during the testing phase results in a further decline in model performance. This further verifies that dataset domain shifts limit the model's restoration capability.
3) To further validate the effectiveness of satellite images, we construct #5, #6, and #7 models, which share the same pre-trained weights but use different reference

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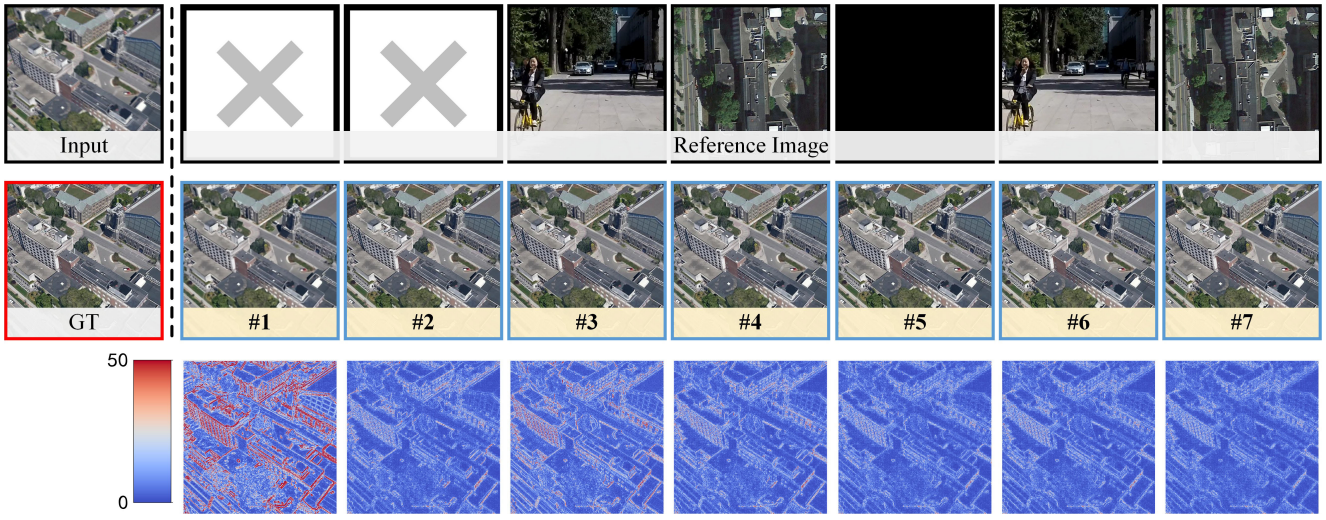


Fig. 15. Deblurred results by methods under different training and testing set configurations.

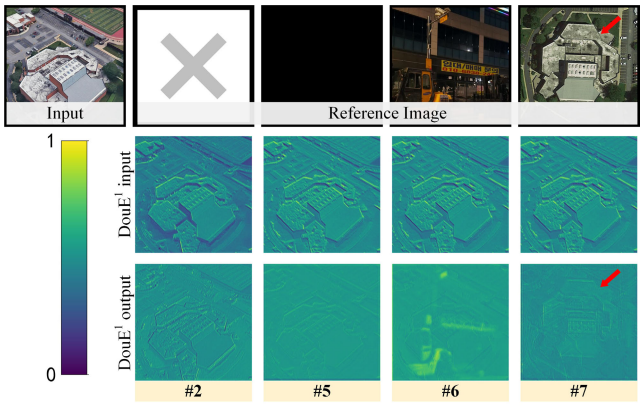


Fig. 16. Heatmaps by methods under different training and testing set configurations.

images during testing. Using a pure black image or a ground-view image as a reference results in an average PSNR decrease of 0.25 and 0.16 dB, respectively. Meanwhile, the #5 model performs slightly worse than the #6 model due to the lack of useful information in the black images.

Fig. 15 presents the comparison of visualization results among the seven models. From the error distribution, the full CVDeblurNet (#7 model) is capable of restoring UAV image details more accurately. In addition, we visualize the input and output features of the DouE^1 in the UAV branch for #5, #6, and #7 models (for the single branch #2 model, the SinE^3 is regarded as the input to DouE^1). As shown in Fig. 16, it can be observed that the structural features of the satellite image are transmitted to the UAV branch and fused with UAV features. We argue that, aided by feature fusion in DouE, satellite structural features enhance the reconstruction quality of UAV images.

2) Impact of Satellite Image Variations: Since the University-1652 dataset covers image data from different countries and regions, satellite images acquired at the same time present distinct variation patterns in different areas. To avoid such interference, we select the subset of Cornell University with the largest number of samples from

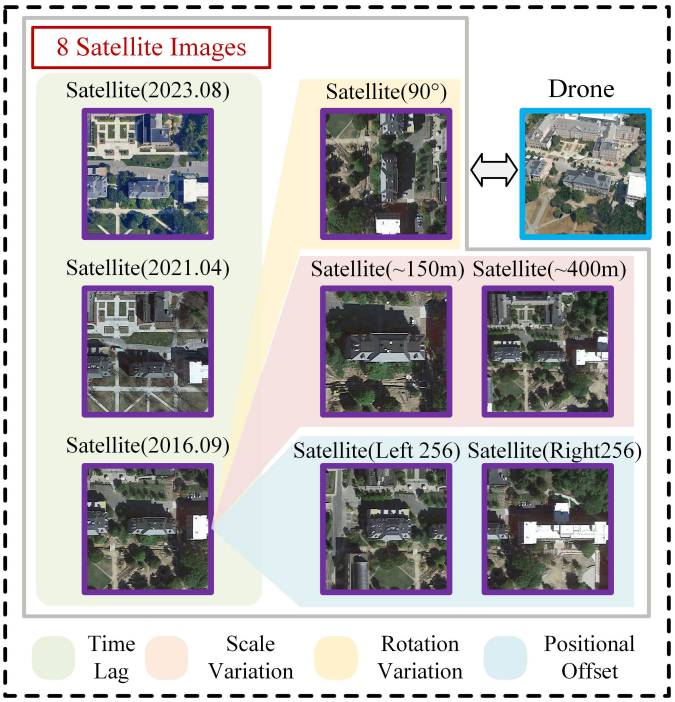


Fig. 17. Visualization of a UAV image together with eight satellite images of different types, including time lag, scale/rotation variation, and position offset.

University-1652 for our experiments, which consists of 119 drone-view images in total. Specifically, we collect satellite images of this area at an approximate imaging altitude of 250 m over three different time periods, denoted as Satellite (2016.09), Satellite (2021.04), and Satellite (2023.08). On the basis of Satellite (2016.09), we further obtain satellite images at approximate imaging altitudes of 150 and 400 m, labeled as Satellite (~150 m) and Satellite (~400 m), respectively. Two new satellite images are generated by shifting Satellite (2016.09) leftward and rightward by 256 pixels each, marked as Satellite(Left 256) and Satellite(Right 256). Additional satellite images are constructed by rotating Satellite (2016.09) clockwise at angles of 90°, marked as Satellite (90°). Fig. 17

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TABLE V

PERFORMANCE COMPARISON OF CVDEBLURNET UNDER DIFFERENT TRAINING AND TESTING SET CONFIGURATIONS, WHERE G, D, S, ANDZDENOTE GROUND-VIEW IMAGES, DRONE-VIEW IMAGES, SATELLITE-VIEW IMAGES, AND PURE BLACK IMAGES, RESPECTIVELY

No.	Training		Testing		Low		Middle		High		Average	
	Main	Ref	Main	Ref	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
#1	G	–	D	–	18.94	0.448	19.26	0.465	19.29	0.462	19.16	0.458
#2	D	–	D	–	28.91	0.921	29.20	0.913	24.43	0.731	27.51	0.855
#3	D	G	D	G	28.25	0.911	29.20	0.914	24.32	0.732	27.26	0.852
#4	D	G	D	S	27.82	0.904	28.93	0.908	24.11	0.720	26.95	0.844
#5 †	D	S	D	Z	29.13	0.921	29.25	0.910	24.14	0.719	27.50	0.850
#6 †	D	S	D	G	29.12	0.922	29.33	0.914	24.32	0.728	27.59	0.855
#7 †	D	S	D	S	**29.28**	**0.926**	**29.53**	**0.918**	24.45	0.731	**27.75**	**0.858**

† denotes #5∼#7 models adopt the same pre-trained weights and different reference images during the testing process.

TABLE VI

PERFORMANCE COMPARISON OF CVDEBLURNET USING VARIOUS SATELLITE IMAGES AS REFERENCES

No.	Satellite Type		Middle	
	Years	Interference	PSNR↑	SSIM↑
1	2016.09	\	31.38	0.9366
2	2021.04	\	31.24	0.9363
3	2023.08	\	31.48	0.9365
4	2016.09	∼150m	31.42	0.9364
5	2016.09	∼400m	31.14	0.9359
6	2016.09	Left offset	31.39	0.9366
7	2016.09	Right offset	31.41	0.9364
8	2016.09	Rotation	31.35	0.9365

TABLE VII

PERFORMANCE COMPARISON OF CVDEBLURNET WITH VARIOUS STRUCTURES

PCLB Structure		Middle		Params (M)
RCB [62]	Enhancement	PSNR↑	SSIM↑	
✓		29.39	0.914	4.22
		28.89	0.906	**3.76**
	✓	**29.53**	**0.918**	3.99

S(D)SKFF Structure		Middle		Params (M)
Add	Select	PSNR↑	SSIM↑	
✓		29.45	0.917	**3.94**
	✓	**29.53**	**0.918**	3.99

MIB Structure			Middle		Params (M)
×1	×2	×4	PSNR↑	SSIM↑	
✓			27.57	0.8801	**0.58**
✓	✓		28.73	0.904	1.61
✓	✓	✓	**29.53**	**0.918**	3.99

shows eight satellite images corresponding to a single UAV image.

We conduct model tests by pairing the UAV images with different types of satellite images as input. Experimental results are summarized in Table VI, and several key conclusions are derived through quantitative comparisons as follows.

1) High-quality reference images contribute more favorably to the performance. By comparing satellite images acquired in three different years, we observe that the 2023 satellite data present appropriate brightness, less shadow occlusion, and noise, whereas the 2021 images suffer from severe noise interference. Accordingly, the testing result based on Satellite (2023.08) is 0.24 dB higher than that using Satellite (2021.04).
2) Satellite images with a visual scale consistent with UAV images are more effective. As the acquisition altitude of satellite images increases, excessive irrelevant information is introduced, and the reference content becomes blurred. For this reason, the model adopting Satellite (∼150 m) achieves superior performance compared with Satellite (∼400 m).
3) Slight positional shift and rotation applied to the default central region of satellite images do cause moderate fluctuations in model performance. Such performance variation is reasonable and acceptable.

3) Effectiveness of Each Module: First, to verify the effectiveness of the PCLB, we remove both its channel and spatial enhancement components for comparative experiments. As shown in Table VII, removing these two components reduces the model's parameter count decreases by 0.23 M while causing a 0.64 dB PSNR drop. Furthermore, replacing the PCLB with the RCB [62] leads to a 0.23 M parameter increase alongside a 0.24 dB decrease in PSNR. These results confirm that the PCLB can improve UAV image reconstruction by establishing global feature dependencies.

Second, we validate the effectiveness of the SSKFB and DSKFB by replacing their feature selection operation with a simple addition operation for fusing multiscale features and cross-view features. As shown in Table VII, the introduction of feature selection operation improves image reconstruction quality with only a 0.05 M parameter increase. This confirms the effectiveness of SSKFB and DSKFB.

Finally, we adjust the number of scales in the MIB to validate the effectiveness of its multiscale structure. Compared with single-scale and dual-scale MIB, the three-scale MIB significantly improves model performance, with PSNR increases of 0.80 and 1.96 dB, respectively.

4) Feature Interaction at Different Stages: To analyze the impact of feature interaction on model performance, we adjust the number and placement of DouE in the drone branch. As shown in Table VIII, compared to other models employing two DouE blocks, our model positions DouE at the third and fourth stages to enhance cross-view feature interaction, achieving superior performance with negligible computational overhead. Notably, later-stage interactions in CVDeblurNet involve more encoders, resulting in a continuous increase in parameter count. Front-end interactions, by

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TABLE VIII
PERFORMANCE COMPARISON OF CVDEBLURNET USING FEATURE INTERACTION AT DIFFERENT STAGES

Stages of interaction				Middle		Params (M)
1	2	3	4	PSNR↑	SSIM↑	
✓				29.08	0.911	**2.49**
	✓			29.05	0.910	2.99
		✓		29.17	0.913	3.49
			✓	29.05	0.910	3.99
	✓	✓		28.96	0.907	3.49
	✓		✓	29.11	0.911	3.99
		✓	✓	**29.53**	**0.918**	3.99

TABLE IX
PERFORMANCE COMPARISON OF CVDEBLURNET WITH DIFFERENT NUMBER OF ENCODERS

Encoder number	Middle		Computational cost	
	PSNR↑	SSIM↑	Params (M)	MACs (G)
2	28.33	0.895	**2.00**	**227.56**
3	28.76	0.905	2.99	340.83
4	**29.53**	**0.918**	3.99	454.10

TABLE X
PERFORMANCE COMPARISON OF CVDEBLURNET WITH DIFFERENT NUMBER OF CHANNELS

Channel	γ	Middle		Computational cost	
		PSNR↑	SSIM↑	Params (M)	MACs (G)
C=16	1	25.97	0.833	**0.18**	**38.58**
C=16	1.5	26.47	0.848	0.44	52.28
C=32	1	28.01	0.889	0.72	149.27
C=32	1.5	28.65	0.903	1.77	203.63
C=48	1	28.69	0.904	1.62	332.10
C=48	1.5	**29.53**	**0.918**	3.99	454.10

contrast, the encoders at the end of the satellite branch can be ignored.

5) Number of Encoders: The depth of CVDeblurNet is determined by the number of encoders, which is identical in its dual branches. As summarized in Table IX, increasing the number of encoders from 2 to 4 leads to a robust performance improvement across all three metrics, albeit with an increase in computational cost. Thus, CVDeblurNet adopts four encoders for each branch by default.

6) Number of Channels: The impact of CVDeblurNet's scale on performance is evaluated by adjusting the base channel number C and expansion factor γ. The experimental results are presented in Table X. Overall, as the channel count of CVDeblurNet increases, both the parameter count and MACs increase, and the model's performance across all metrics also improves accordingly. Thus, the default values of C and γ in CVDeblurNet are set to 48 and 1.5, respectively.

VI. CONCLUSION

In this article, we propose a satellite-guided UAV image deblurring task and construct the CVDeblur dataset. The CVDeblur dataset consists of a training set, a synthetic test set, and a real test set. The former two sets contain 6298 triplets, where each triplet comprises a blurred UAV image, a sharp UAV image, and a geographically consistent satellite image. The real test set contains 120 pairs of blurred UAV images and satellite images. Subsequently, we design the CVDeblurNet, a dual-branch network that takes both UAV and satellite images as inputs and adaptively integrates satellite features into UAV features to compensate for the lost details. Experimental results demonstrate that CVDeblurNet can extract beneficial information from satellite images and improve the quality of UAV image reconstruction.

REFERENCES

[1] Q. Li, Y. Zhang, L. Fang, Y. Kang, S. Li, and X. X. Zhu, "DREB-Net: Dual-stream restoration embedding blur-feature fusion network for high-mobility UAV object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5621218.

[2] J. Wang, P. Du, S. Yang, Z. Zhang, and J. Ning, "A spatial arrangement preservation-based stitching method via geographic coordinates of UAV for farmland remote sensing image," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4404713.

[3] Q. Chen et al., "SDPL: Shifting-dense partition learning for UAV-view geo-localization," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 11, pp. 11810–11824, Nov. 2024.

[4] Q. Chen, T. Wang, R. Lu, Y. Liu, B. Zheng, and Z. Zheng, "Scale-adaptive UAV geo-localization via height-aware partition learning," 2024, *arXiv:2412.11535*.

[5] J. Liang et al., "Ntire 2025 the 2nd restore any image model (RAIM) in the wild challenge," in *Proc. Comput. Vis. Pattern Recognit. Conf.*, Aug. 2025, pp. 1269–1278.

[6] Z. Yang, Y. Sun, X. Peng, S. M. Yiu, and Y. Ma, "Unidemoiré: Towards universal image demoiréing with data generation and synthesis," in *Proc. AAAI Conf. Artif. Intell.*, 2025, vol. 39, no. 9, pp. 9354–9362.

[7] J. Jiang, Z. Zuo, G. Wu, K. Jiang, and X. Liu, "A survey on all-in-one image restoration: Taxonomy, evaluation and future trends," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 13, pp. 11892–11911, Dec. 2025.

[8] W. Lu, J. Wang, T. Wang, K. Zhang, X. Jiang, and H. Zhao, "Visual style prompt learning using diffusion models for blind face restoration," *Pattern Recognit.*, vol. 161, Jul. 2025, Art. no. 111312.

[9] K. Zhang, W. Luo, Y. Zhong, L. Ma, W. Liu, and H. Li, "Adversarial spatio-temporal learning for video deblurring," *IEEE Trans. Image Process.*, vol. 28, no. 1, pp. 291–301, Jan. 2019.

[10] D. Li, Y. Zhang, K. C. Cheung, X. Wang, H. Qin, and H. Li, "Learning degradation representations for image deblurring," in *Proc. Eur. Conf. Comput. Vis.*, Oct. 2022, pp. 736–753.

[11] F. Ji, J. Wang, S. Cui, J. Li, X. Tang, and F. Xu, "Physics-guided optical simulation and PSF analysis for remote sensing images deblurring," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5633015.

[12] Y. Feng, Y. Yang, X. Fan, Z. Zhang, and J. Zhang, "A multiscale generalized shrinkage threshold network for image blind deblurring in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611316.

[13] H. Liu, J. Sun, J. Liu, X.-L. Zhao, T. Wu, and T. Zeng, "Blind noisy image deblurring using residual guidance strategy," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Sep. 2025, pp. 11016–11025.

[14] H. Yang, J. Rim, S. Lee, S.-H. Baek, and S. Cho, "Gyro-based neural single image deblurring," in *Proc. Comput. Vis. Pattern Recognit. Conf.*, 2025, pp. 23111–23120.

[15] Z. Fang et al., "Parameterized blur kernel prior learning for local motion deblurring," in *Proc. Comput. Vis. Pattern Recognit. Conf.*, 2025, pp. 23006–23015.

[16] L. Tu, J. Wu, C. Wang, D. Meng, and Z. Jin, "Fourier-based decoupling network for joint low-light image enhancement and deblurring," *IEEE Trans. Image Process.*, vol. 34, pp. 5184–5199, Jul. 2025.

[17] K. Zhang et al., "Deep image deblurring: A survey," *Int. J. Comput. Vis.*, vol. 130, no. 9, pp. 2103–2130, Sep. 2022.

[18] S. Nah, T. H. Kim, and K. M. Lee, "Deep multi-scale convolutional neural network for dynamic scene deblurring," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3883–3891.

[19] K. Zhang et al., "MC-blur: A comprehensive benchmark for image deblurring," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 5, pp. 3755–3767, May 2024.

[20] R. Fergus, B. Singh, A. Hertzmann, S. T. Roweis, and W. T. Freeman, "Removing camera shake from a single photograph," in *Proc. ACM SIGGRAPH Papers (SIGGRAPH)*, 2006, pp. 787–794.

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[21] A. Levin, Y. Weiss, F. Durand, and W. T. Freeman, "Understanding and evaluating blind deconvolution algorithms," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 1964–1971.

[22] L. Xu, S. Zheng, and J. Jia, "Unnatural l0 sparse representation for natural image deblurring," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2013, pp. 1107–1114.

[23] A. Levin, Y. Weiss, F. Durand, and W. T. Freeman, "Efficient marginal likelihood optimization in blind deconvolution," in *Proc. CVPR*, Jun. 2011, pp. 2657–2664.

[24] J. Pan, D. Sun, H. Pfister, and M.-H. Yang, "Blind image deblurring using dark channel prior," in *Proc. IEEE Conf. Comput. Vis. pattern Recognit.*, Aug. 2016, pp. 1628–1636.

[25] O. Kupyn, V. Budzan, M. Mykhailych, D. Mishkin, and J. Matas, "DeblurGAN: Blind motion deblurring using conditional adversarial networks," in *Proc. IEEE Conf. Comput. Vis. pattern Recognit.*, Aug. 2018, pp. 8183–8192.

[26] I. Goodfellow et al., "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.

[27] X. Tao, H. Gao, X. Shen, J. Wang, and J. Jia, "Scale-recurrent network for deep image deblurring," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Aug. 2018, pp. 8174–8182.

[28] J. Kim, J. K. Lee, and K. M. Lee, "Deeply-recursive convolutional network for image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2016, pp. 1637–1645.

[29] K. Zhang et al., "Deblurring by realistic blurring," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 2734–2743.

[30] S. Nah et al., "NTIRE 2019 challenge on video deblurring and super-resolution: Dataset and study," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2019, pp. 1996–2005.

[31] Z. Shen et al., "Human-aware motion deblurring," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Jun. 2019, pp. 5572–5581.

[32] Y. Zhang, P. Zheng, C. Zeng, B. Xiao, Z. Li, and X. Gao, "Jointly RS image deblurring and super-resolution with adjustable-kernel and multi-domain attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2024, Art. no. 4700416.

[33] X. Yang, X. Wang, N. Wang, and X. Gao, "SRDN: A unified super-resolution and motion deblurring network for space image restoration," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022.

[34] J.-H. Lee, G.-H. Gwon, I.-H. Kim, and H.-J. Jung, "A motion deblurring network for enhancing uav image quality in bridge inspection," *Drones*, vol. 7, no. 11, p. 657, 2023.

[35] C. Guo, Y. Ji, Y. Liu, H. Zheng, J. Li, and H. Cao, "Restoration of uav image blurring based on inertial measurement unit," *Meas. Sci. Technol.*, vol. 36, no. 1, 2024, Art. no. 015137.

[36] V. Sineglazov, K. Lesohorskyi, and O. Chumachenko, "Faster image deblurring for unmanned aerial vehicles," in *Proc. 2nd Int. Conf. Unmanned Vehicle Syst.-Oman (UVS)*, Oct. 2024, pp. 1–6.

[37] Y. Zhang, K. Li, K. Li, L. Wang, B. Zhong, and Y. Fu, "Image super-resolution using very deep residual channel attention networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 286–301.

[38] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient channel attention for deep convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11534–11542.

[39] Z. Qin, P. Zhang, F. Wu, and X. Li, "FcaNet: Frequency channel attention networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Jul. 2021, pp. 783–792.

[40] X. Tong et al., "GSFNet: Gyro-aided spatial-frequency network for motion deblurring of uav infrared images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5003718.

[41] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, Nov. 2004.

[42] C. Barnes, E. Shechtman, A. Finkelstein, and D. B. Goldman, "Patchmatch: A randomized correspondence algorithm for structural image editing," *ACM Trans. Graph.*, vol. 28, no. 3, p. 24, 2009.

[43] M. Zhao, G. Wu, Y. Li, X. Hao, L. Fang, and Y. Liu, "Cross-scale reference-based light field super-resolution," *IEEE Trans. Comput. Imag.*, vol. 4, no. 3, pp. 406–418, Sep. 2018.

[44] G. Shim, J. Park, and I. S. Kweon, "Robust reference-based super-resolution with similarity-aware deformable convolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Apr. 2020, pp. 8425–8434.

[45] R. Dong, L. Zhang, and H. Fu, "RRSGAN: Reference-based super-resolution for remote sensing image," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.

[46] Y. Jiang, K. C. Chan, X. Wang, C. C. Loy, and Z. Liu, "Reference-based image and video super-resolution via C^2-matching," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 7, pp. 8874–8887, Jul. 2022.

[47] H. Zheng, M. Ji, H. Wang, Y. Liu, and L. Fang, "CrossNet: An end-to-end reference-based super resolution network using cross-scale warping," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 88–104.

[48] A. Dosovitskiy et al., "FlowNet: Learning optical flow with convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 2758–2766.

[49] Y. Jiang, K. C. K. Chan, X. Wang, C. C. Loy, and Z. Liu, "Robust reference-based super-resolution via C2-matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2103–2112.

[50] Y. Zhang, Q. Yang, D. M. Chandler, and X. Mou, "Reference-based multi-stage progressive restoration for multi-degraded images," *IEEE Trans. Image Process.*, vol. 33, pp. 4982–4997, Sep. 2024.

[51] M. Bernasconi, A. Djelouah, Y. Zhang, M. Gross, and C. Schroers, "Rebair: Reference-based image restoration," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Sep. 2025, pp. 5489–5498.

[52] J. Yan et al., "RASR: Retrieval-augmented super resolution for practical reference-based image restoration," 2025, *arXiv:2508.09449*.

[53] M. Zhou et al., "Reference-guided identity preserving face restoration," 2025, *arXiv:2505.21905*.

[54] X. Luo, J. Peng, W. Zhao, K. Xian, H. Lu, and Z. Cao, "Point-and-shoot all-in-focus photo synthesis from smartphone camera pair," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 5, pp. 2088–2101, May 2023.

[55] Z. Tu, X. Yang, X. He, J. Yan, and T. Xu, "RGTGAN: Reference-based gradient-assisted texture-enhancement GAN for remote sensing super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5607221.

[56] J. Rim, H. Lee, J. Won, and S. Cho, "Real-world blur dataset for learning and benchmarking deblurring algorithms," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2020, pp. 184–201.

[57] J. Rim, G. Kim, J. Kim, J. Lee, S. Lee, and S. Cho, "Realistic blur synthesis for learning image deblurring," in *Proc. Eur. Conf. Comput. Vis. Cham*, Switzerland: Springer, 2022, pp. 487–503.

[58] H. Li, Z. Zhang, T. Jiang, P. Luo, H. Feng, and Z. Xu, "Real-world deep local motion deblurring," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, no. 1, pp. 1314–1322.

[59] K. Lee, Y. Ban, and C. Kim, "Motion blur kernel rendering using an inertial sensor: Interpreting the mechanism of a thermal detector," *Sensors*, vol. 22, no. 5, p. 1893, Feb. 2022.

[60] K. Ko, K. Shim, K. Lee, and C. Kim, "Large-scale benchmark for uncooled infrared image deblurring," *IEEE Sensors J.*, vol. 23, no. 24, pp. 30119–30128, Dec. 2023.

[61] Z. Zheng, Y. Wei, and Y. Yang, "University-1652: A multi-view multi-source benchmark for drone-based geo-localization," *ACM Multimedia*, pp. 1395–1403, Oct. 2020.

[62] S. W. Zamir et al., "Learning enriched features for fast image restoration and enhancement," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 1934–1948, Feb. 2023.

[63] X. Li, W. Wang, X. Hu, and J. Yang, "Selective kernel networks," in *Proc. IEEE/CVF Conf. Comput. Vis. pattern Recognit.*, Aug. 2019, pp. 510–519.

[64] W. Shang, D. Ren, W. Zhang, P. Zhu, Q. Hu, and W. Zuo, "Motion-aware adaptive pixel pruning for efficient local motion deblurring," in *Proc. 33rd ACM Int. Conf. Multimedia*, 2025, pp. 7700–7708.

[65] B. Zhang, J. Sun, F. Sun, F. Wang, and B. Zhu, "Image deblurring method based on self-attention and residual wavelet transform," *Expert Syst. Appl.*, vol. 244, Jul. 2024, Art. no. 123005.

[66] H. Liu, C. Liu, J. Xu, P. Jiang, and M. Lu, "Xyscannet: A state space model for single image deblurring," in *Proc. Comput. Vis. Pattern Recognit. Conf.*, Apr. 2025, pp. 779–789.

[67] L. Kong, J. Dong, J. Tang, M.-H. Yang, and J. Pan, "Efficient visual state space model for image deblurring," in *Proc. Comput. Vis. Pattern Recognit. Conf.*, Jul. 2025, pp. 12710–12719.

[68] L. Kong, J. Dong, J. Ge, M. Li, and J. Pan, "Efficient frequency domain-based transformers for high-quality image deblurring," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Aug. 2023, pp. 5886–5895.

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[69] S. W. Zamir et al., "Learning enriched features for real image restoration and enhancement," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2020, pp. 492–511.

[70] S. W. Zamir, A. Arora, S. Khan, M. Hayat, F. S. Khan, and M.-H. Yang, "Restormer: Efficient transformer for high-resolution image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5718–5729.

[71] P.-H. Kuo, J. Pan, S.-Y. Chien, and M.-H. Yang, "Efficient concertormer for image deblurring and beyond," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2025, pp. 14665–14675.

[72] X. Mao, J. Wang, X. Xie, Q. Li, and Y. Wang, "Loformer: Local frequency transformer for image deblurring," in *Proc. 32nd ACM Int. Conf. Multimedia*, 2024, pp. 10382–10391.

[73] Q.-T. Ho, M.-T. Duong, S. Lee, and M.-C. Hong, "EHNet: Efficient hybrid network with dual attention for image deblurring," *Sensors*, vol. 24, no. 20, p. 6545, Oct. 2024.

[74] Y. Qiu, K. Zhang, C. Wang, W. Luo, H. Li, and Z. Jin, "Mb-Taylorformer: Multi-branch efficient transformer expanded by Taylor formula for image dehazing," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 12802–12813.

[75] Z. Jin, Y. Qiu, K. Zhang, H. Li, and W. Luo, "MB-TaylorFormer v2: Improved multi-branch linear transformer expanded by Taylor formula for image restoration," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 7, pp. 5990–6005, Jul. 2025.

[76] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent*., vol. 9351. Cham, Switzerland: Springer, 2015, pp. 234–241.

[77] L. Chen, X. Chu, X. Zhang, and J. Sun, "Simple baselines for image restoration," in *Proc. Eur. Conf. Comput. Vis. Cham*, Switzerland: Springer, 2022, pp. 17–33.

[78] S. W. Zamir et al., "Multi-stage progressive image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Aug. 2021, pp. 14821–14831.

[79] F.-J. Tsai, Y.-T. Peng, Y.-Y. Lin, C.-C. Tsai, and C.-W. Lin, "Stripformer: Strip transformer for fast image deblurring," in *Proc. Eur. Conf. Comput. Vis. Cham*, Switzerland: Springer, 2022, pp. 146–162.

[80] X. Gao et al., "Efficient multi-scale network with learnable discrete wavelet transform for blind motion deblurring," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Sep. 2024, pp. 2733–2742.

[81] I. Q. Assessment, "From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, p. 93, Apr. 2004.

[82] I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts," 2016, *arXiv:1608.03983*.

[83] D. P. Kingma, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[84] J. Ke, Q. Wang, Y. Wang, P. Milanfar, and F. Yang, "Musiq: Multi-scale image quality transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Jul. 2021, pp. 5148–5157.

[85] S. Yang et al., "Maniqa: Multi-dimension attention network for no-reference image quality assessment," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Apr. 2022, pp. 1191–1200.

[86] J. Wang, K. C. Chan, and C. C. Loy, "Exploring clip for assessing the look and feel of images," in *Proc. AAAI Conf. Artif. Intell.*, vol. 37, no. 2, 2023, pp. 2555–2563.

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